

Intelligent Engine for Financial Forecasting

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Quick Access

Live application	https://fincast--simonsibanda26.replit.app
Github Repo	https://github.com/makoesibanda/fincast
Demo Video	https://www.youtube.com/watch?v=in36PqbS_rQ
Admin User	fincast
Admin Password	admin1234

What the software does

I developed and named this application '**FinCast**'. It is a full-stack web application that predicts the next-day Bitcoin price direction. It uses a hybrid model that combines three approaches: GRU, Transformer, and Linear Regression. Their outputs are blended using weights found during validation (GRU=0.7, Transformer=0.3, Linear Regression=0.0).

The system is built to be usable, not just accurate. The dashboard explains each prediction in simple terms, while the Models page shows full evaluation results. There is also a Sentiment page where live financial news is scored using FinBERT.

The main idea behind the project was to bring multiple components into one system. Many existing approaches either focus only on prediction accuracy or ignore explainability and real-world usability. In this project, the aim was to combine forecasting, sentiment analysis, evaluation metrics, and a working interface into a single pipeline.

The model was tested on data it had never seen before, and the system is designed so both technical and non-technical users can understand what the prediction means.

Live Demo

The application is deployed and accessible at, <https://fincast--simonsibanda26.replit.app>

Note: The app is hosted on Replit's free tier which spins down after inactivity. If the app is slow or shows an error on first load, the server may be waking up. Wait a few seconds and try again.

Core Features

The system provides the following functionality across the application:

- User authentication with secure password hashing and JWT-based session handling

- Next-day BTC-USD direction prediction using the hybrid fusion model (GRU + Transformer)
- Live sentiment analysis using FinBERT on financial news headlines from RSS feeds
- Interactive dashboard showing:
 - predicted direction
 - confidence level
 - volatility (14-day)
 - sentiment score
- Plain-language explanation of each prediction so users can understand what the model is suggesting
- Confidence gauge with visual feedback to indicate prediction strength
- Historical prediction tracking stored per user in the database
- 60-day Bitcoin price chart rendered on the dashboard
- SHAP-based feature importance showing which inputs influence predictions the most
- Models page displaying full evaluation results, including:
 - Sharpe ratio
 - directional accuracy
 - maximum drawdown
 - walk-forward validation
- Sentiment page displaying live scored headlines with individual sentiment values
- Admin panel allowing:
 - user management (activate / deactivate)
 - role control
 - platform activity monitoring
- Currency conversion feature supporting multiple currencies for price interpretation

Model Development and Training

Note on Reproducibility:

The submitted notebook was rerun on a different machine after the original laptop failed and the initial outputs were lost. Due to the random initialisation of neural network weights in PyTorch, some results from the rerun differ slightly from the original training run.

The results reported in this document, as well as those shown in the live application, are based on the original training outputs.

The trained model files were recovered from a cloud backup and are the same ones currently used by the deployed system.

The majority of the project time was spent on model development in the Jupyter notebook. The model was developed following a structured workflow, starting from data collection through to final evaluation. The aim

was to build a system that can predict next-day price direction while remaining interpretable and stable.

The **dataset** combines historical Bitcoin price data with engineered technical indicators. Price data was collected using the **yfinance API** and covers 2018 to 2024. **The split was training on 2018-2022, validation on 2023, and a held-out test set on 2024.** The data includes open, high, low, close, and volume values. From this, multiple features were created, including returns, moving averages, RSI, MACD, volatility measures, and volume-based indicators. All features were designed to match the configuration used during training to avoid inconsistencies at inference time.

Sentiment was introduced as an additional feature using FinBERT. Since historical sentiment data was not available for the full training period, a zero placeholder was used during training. In the live system, this is replaced with real-time sentiment scores derived from financial news headlines.

The model uses a sliding window approach, where the previous 30 days of data are used to predict the next day. Feature scaling is applied using scalers fitted only on training data to prevent data leakage.

Three models were used:

- A GRU model, which captures sequential patterns in time series data
- A Transformer model, which captures longer-range dependencies using attention
- A Linear Regression model, used as a baseline

Each deep learning model produces two outputs: a regression value for expected return and a classification probability for direction. These outputs are combined using a weighted fusion approach.

The final prediction is generated using the **Hybrid Fusion Model (HFM)**:

- GRU weight: 0.7
- Transformer weight: 0.3
- Linear Regression weight: 0.0

The Linear Regression model was excluded from the final fusion as it did not improve performance during validation.

On the 2024 held-out test set, GRU achieved a Sharpe ratio of 2.612 and directional accuracy of 54.19%. The HFM ensemble achieved Sharpe 2.209 and directional accuracy of 52.40% with a maximum drawdown of -18.1%. Walk-forward validation across three 90-day windows confirmed the model generalises beyond the training period, though performance dropped in Window 2 during the mid-2024 Bitcoin consolidation period.

SHAP GradientExplainer was used to identify which features influenced predictions most. Return_14, MACD_Signal and MACD_Hist ranked highest. Sentiment ranked last because it was zero during training.

The predicted return is converted into a predicted price using the latest market price. The direction is determined based on the classification output, and confidence is derived from the probability associated with that direction.

To improve stability, predicted returns are capped within a fixed range, and inconsistencies between regression and classification outputs are corrected before returning the final result.

Access and Login

Users can register a new account either through the live application or when running the system locally.

There are no pre-created user accounts. Each user must register before accessing the dashboard.

For administrative access, a default admin account is automatically created when the application is first started. The default credentials are:

- **Username:** fincast
- **Password:** admin1234

After logging in with admin credentials, the system redirects to the admin panel, where user accounts and platform activity can be managed

Local Setup (*tested several times from my Windows PC*)

Requirements

Local Setup (Windows)

Important: Python 3.12 is required. Python 3.13 and 3.14 are not compatible. Download Python 3.12.13 from: <https://www.python.org/downloads/release/python-31213/>

During installation, tick **Add Python to PATH**.

Requirements

- Python 3.12.13
- Internet connection (required for live price data, sentiment feeds and currency conversion)
- Git

Step 1 - Clone the repository

```
git clone https://github.com/makoesibanda/fincast
```

```
cd fincast
```

Step 2 - Navigate to the backend

```
cd backend
```

Step 3 - Create a virtual environment

```
python -m venv venv
```

Step 4 - Activate the virtual environment

```
venv\Scripts\activate
```

You should see (venv) at the start of your terminal line.

If you get a scripts disabled error, run this first then try again:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Step 5 - Install dependencies

```
pip install -r ../requirements.txt
```

Step 6 - Run the application

```
python -m flask --app app run --no-reload
```

Step 7 - Open in browser

<http://127.0.0.1:5000>

Technology Stack

The system was built using the following technologies across different layers:

LAYER	TECHNOLOGY
Backend	Python 3.12, Flask 3.0, SQLAlchemy
ML Models	PyTorch (GRU, Transformer), scikit-learn (Linear Regression, Random Forest), XGBoost
Model Training	Adam optimiser, learning rate scheduler, early stopping, MinMaxScaler, StandardScaler
Explainability	SHAP GradientExplainer on 80 test samples
NLP / Sentiment	FinBERT (ProsusAI/finbert) via HuggingFace Transformers
Data	yfinance, pandas, numpy - 19 features from BTC-USD OHLCV data
Notebook charts	matplotlib, seaborn
Database	SQLite
Authentication	JWT, Werkzeug password hashing
Currency API	open.er-api.com
Frontend	HTML, CSS, vanilla JavaScript, Chart.js
Deployment	Replit (upgrading to paid tier before marking period ends)

Known Limitations

Only BTC-USD is Live

The market selection page lists Cryptocurrencies, Stocks and Forex as categories. Within Cryptocurrencies, five assets are shown: BTC-USD, ETH-USD, XRP-USD, LTC-USD and ADA-USD. Currently, only BTC-USD generates live predictions. The remaining four show a coming soon message.

The significant time investment required to converge on a stable, well-performing model for BTC-USD, including feature engineering, hyperparameter tuning, walk-forward validation and ensemble weight optimisation, meant that extending the pipeline to additional assets was not feasible within the project timeline.

Bitcoin was prioritised because it has the longest continuous price history of any cryptocurrency, the highest liquidity, and the richest available data, making it the strongest candidate for a proof-of-concept deep learning forecasting system.

The multi-asset interface was designed and built to reflect the intended scope of the system, and the architecture supports additional assets without major structural changes.

Sentiment Analysis - Training Data Limitation

Sentiment was identified in the project proposal and literature review as a core component of the system, addressing the gap around models that rely solely on price data and ignore the market's reaction to news.

FinBERT was successfully integrated, and the live pipeline is fully operational. Headlines are fetched from financial RSS feeds, scored in real time, and fed into the model as the 19th feature at inference time.

However, during training, the sentiment feature was a zero placeholder throughout the entire 2018 to 2022 period. This was because historical financial news archives with timestamps matching daily BTC price data are not freely available. As a result, the model never learned any relationship between sentiment scores and price movements.

SHAP analysis confirms this, with sentiment ranked last out of 19 features and showing a near-zero mean absolute SHAP value.

The live FinBERT pipeline demonstrates the intended architecture, and the sentiment score is visible to users on both the dashboard and the sentiment page. However, its actual influence on predictions is effectively zero until the model is retrained on data that includes real historical sentiment values.

SHAP Explanations are Static

SHAP was computed in the notebook on 80 test samples. Results are displayed statically and not recalculated per prediction. Running SHAP at inference time would add significant delay to every request, and since SHAP values reflect the model's general behaviour across many samples rather than any single prediction, computing it once on the test set is consistent with how it is used in academic research.

SHAP results do not change unless the model is retrained.

Predictions are Daily

FinCast is a daily forecasting system. It predicts the next-day price direction based on the most recent 30 days of market data.

Running the prediction multiple times within the same day will produce near-identical results because the 30-day input window changes only marginally between requests.

The model is effectively seeing the same historical data until a new daily candle closes. This is expected behaviour, not a bug.

Rolling Retraining Not Implemented

The dataset was defined as 2018 to 2024 at the outset of the project. In hindsight, extending the data to include 2025, retraining on 2018 to 2023, validating on 2024 and testing on 2025 would have been a stronger experimental design.

This was not done because the full pipeline, including feature engineering, model training, hyperparameter tuning and ensemble optimisation, **consumed the available project time before this gap was identified.**

Updating the dataset and implementing a rolling retraining pipeline remains the most important next step for future development.

Hosting - Goldsmiths Server and Replit

Deployment was first attempted on the Goldsmiths university server. When attempting to install the required version of PyTorch, the server returned: "Could not find a version that satisfies the requirement torch>=2.2.0. No matching distribution found."

```

ssiba001@doc350:~$ pip3 install "torch>=2.2.0"
Collecting torch>=2.2.0
  Could not find a version that satisfies the requirement torch>=2.2.0 (from versions: 1.0.0, 1.0.1, 1.1.0, 1.2.0, 1.3.0, 1.3.1, 1.4.0, 1.5.0, 1.5.1, 1.6.0, 1.7.0, 1.7.1, 1.8.0, 1.8.1, 1.9.0, 1.9.1, 1.10.0, 1.10.1, 1.10.2)
No matching distribution found for torch>=2.2.0

```

The project requires torch 2.2.0 or higher, which is the version under which the GRU and Transformer models were trained and saved. The server is therefore incompatible with this project.

Replit was selected as the alternative deployment platform. It runs Python 3.12.12 with PyTorch 2.11.0 installed, fully compatible with all project dependencies and model files.

Replit's free tier spins down after a period of inactivity. When the server is waking up, the first request may fail with a network or JSON error. Waiting a few seconds and trying again resolves this. A notice is displayed at the bottom of every page to inform users of this behaviour.

As the marking period is unknown, the plan is to upgrade to a paid tier before the free trial expires to ensure the application remains accessible throughout the assessment period. If the application is unavailable at the time of marking, the local setup instructions provide a fully working alternative

Future Work

The current system provides a complete end-to-end forecasting pipeline, but several improvements can be made.

A key priority is incorporating historical sentiment data during training. In the current system, sentiment is only used at inference time, which limits its impact on predictions. Integrating time-aligned historical news data would allow the model to learn how market sentiment influences price movements, making this the most important improvement for future development.

The system can also be extended to support multiple assets. While the interface includes cryptocurrencies, stocks, and forex, only BTC-USD is currently active.

Another improvement is implementing a rolling retraining pipeline so the model can adapt to new market data over time.

Finally, deployment can be improved by using a more stable hosting environment to remove cold start delays and improve reliability.

References

- [1] Sezer, O.B., Gudelek, M.U. and Ozbayoglu, A.M. "Financial time series forecasting with deep learning: A systematic literature review". In: Applied Soft Computing 90 (2020), p. 106181.
- [2] Fischer, T. and Krauss, C. "Deep learning with long short-term memory networks for financial market predictions". In: European Journal of Operational Research 270.2 (2018), pp. 654–669.
- [3] Vaswani, A. et al. "Attention Is All You Need". In: Advances in Neural Information Processing Systems 30 (2017).
- [4] Cho, K. et al. "Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation". In: arXiv:1406.1078 (2014).
- [5] Araci, D. "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models". In: arXiv:1908.10063 (2019).
- [6] Lundberg, S.M. and Lee, S.I. "A Unified Approach to Interpreting Model Predictions". In: Advances in Neural Information Processing Systems 30 (2017), pp. 4765–4774.
- [7] Sharpe, W.F. "The Sharpe Ratio". In: Journal of Portfolio Management 21.1 (1994), pp. 49–58.
- [8] Chen, T. and Guestrin, C. "XGBoost: A Scalable Tree Boosting System". In: Proceedings of the

22nd ACM SIGKDD Conference (2016), pp. 785–794.

- [9] Paszke, A. et al. "PyTorch: An Imperative Style, High-Performance Deep Learning Library". In: Advances in Neural Information Processing Systems 32 (2019).
- [10] Nakamoto, S. "Bitcoin: A Peer-to-Peer Electronic Cash System". 2008. Available at: <https://bitcoin.org/bitcoin.pdf>